

Final Project: Build a story!

0. Pick a cancer, not a specific dataset!

Questions to address in the project (at least 3 should be answered):

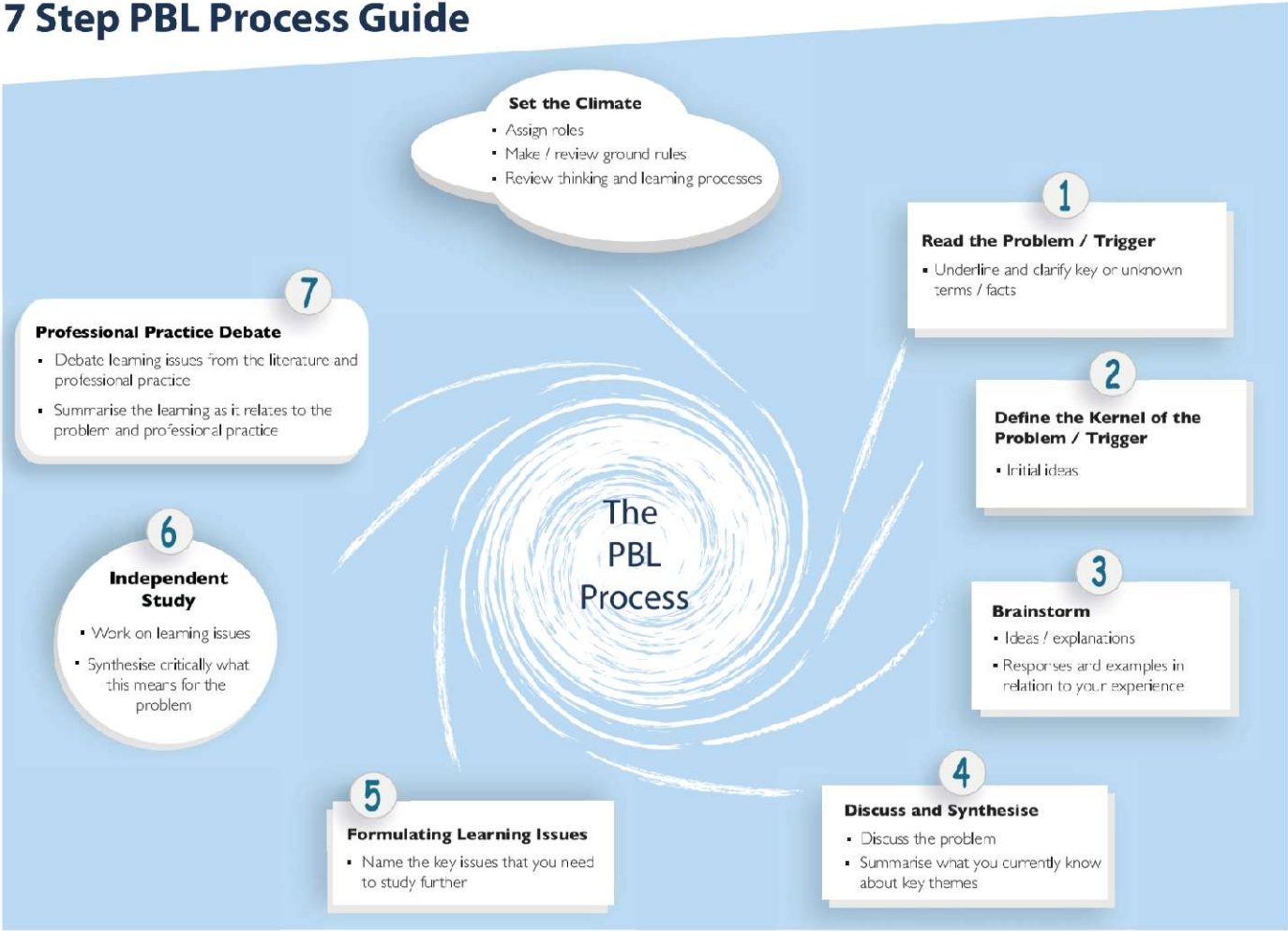
1. Can you build a predictor of patient response, which works across different datasets? You can play with a compromise between data availability and data completeness.
2. Can you build a predictor of cell viability with respect to the drug treatments which are now standard of care? Does it in any way correlate with p.1?
3. Are there any ODE models that are useful?
4. Can you suggest potential new drug targets using LINCS data? Or based on DepMap data? Or both?
5. Can you try to suggest which type of treatment should be used when multiple options are available (e.g. chemotherapy vs immunotherapy)
6. You own creativity.

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- 1. I suggest 4 to 6 persons per group**
- 2. I give you 2 weeks for self-organization**
- 3. Anyone who is not in the group after 2 weeks will be assigned by lecturer**
- 4. If author contribution statement is added in the final presentation, I will grade according to the contribution, if not, everyone in the group will receive equal score.**

Problem based learning guide for the group work

7 Step PBL Process Guide



Roles of individuals in the group

- Chairperson
- Scribe
- Reader
- Timekeeper

The role of the tutor is to:

- Facilitate the PBL process, not to give a mini-lecture
- Listen very attentively to what students are saying and the learning that is taking place in the team
- Encourage a welcoming and challenging learning climate
- Ask questions that encourage critical thinking
- Encourage student to link theory and practice
- Encourage students to be responsible to complete high quality independent learning
- Facilitate students to reflect on their learning, the development of key skills and the performance of the team
- Facilitate the review section of the tutorial

Students all work on the problem and in addition some students take on the roles of: chairperson, scribe, reader and timekeeper.

The role of the chairperson is to:

- Encourage the participation of all team members
- Not necessarily to talk first and certainly not to talk at length
- Facilitate the team to make and work within agreed ground rules
- Stimulate the debate by encouraging discussion of different viewpoints and asking questions
- Use the PBL process as a scaffold for the team to work on the problem
- Ensure that someone summarises at the end of a tutorial
- Check that everyone is clear what learning issues the team has decided to work on.

The role of the scribe is to:

- Record the ideas of the team on the whiteboard so that this can be used as a shared learning environment

- Write the learning issues that the team decide to work on clearly
- Work both verbally and visually on the whiteboard and invite other team members to write on the whiteboard if they want to illustrate a point
- Summarise and synthesise the learning from the problem on the whiteboard

The role of the reader is to:

- Read the problem aloud at the start of the process
- Re-read the problem again when the team decide this is useful
- Draw the team's attention to key elements of the problem

The role of the timekeeper is to:

- Help the team to manage their time in tutorials
- Remind the team appropriately about how much time they have left in the tutorial

In different contexts it is decided to use other roles e.g. observer, photographer or roles that mirror specific professional situations.

In PBL students are often given a PBL tutorial process guide of steps to act as a scaffold for learning. Page 9 shows a PBL process guide used by students on a masters programme in ultrasound in UCD. PBL itself needs to be problematised, adapted and re-invented for different contexts and professions.

The best way to learn PBL is to do it!